7CS039 Lab 4 Testing

1 Objectives

By the end of this session we will be able to

- Perform significance testing using `R`.
- Interpret results from significance tests.
- Perform test for normality using `R`.

2 Birth Weights

We are going to be using the `birthwt` data set today. It's part of the `MASS` package in `R`. We load it into `R` as usual and preview the data.

```
> library(MASS)
> head(birthwt)
```

```
> library(MASS)
> head(birthwt)
   low age lwt race smoke ptl ht ui ftv  bwt
85   0  19 182    2     0   0  0  1   0 2523
86   0  33 155    3     0   0  0  0   3 2551
87   0  20 105    1     1   0  0  0   1 2557
88   0  21 108    1     1   0  0  1   2 2594
89   0  18 107    1     1   0  0  1   0 2600
91   0  21 124    3     0   0  0  0   0 2622
>
```

The `birthwt` data frame has 189 rows and 10 columns. The data were collected at Baystate Medical Center, Springfield, Mass during 1986. This data frame contains the following columns

- low - indicator of birth weight less than 2.5 kg.
- age - mother's age in years.
- lwt - mother's weight in pounds at last menstrual period.

- race - mother's race (1 = white, 2 = black, 3 = other).
- smoke - smoking status during pregnancy.
- ptl - number of previous premature labours.
- ht - history of hypertension.
- ui- presence of uterine irritability.
- ftv - number of physician visits during the first trimester.
- bwt- birth weight in grams.

We'll be using some functions from the `tidyverse` package so it's a good idea to load this now.

```
> library(tidyverse)
```

```
> library(tidyverse)
── Attaching packages ─────────────── tidyverse 1.3.0 ──
✓ ggplot2 3.3.2     ✓ purrr   0.3.4
✓ tibble  3.0.4     ✓ dplyr   1.0.2
✓ tidyr   1.1.2     ✓ stringr 1.4.0
✓ readr   1.4.0     ✓ forcats 0.5.0
── Conflicts ────────────────── tidyverse_conflicts() ──
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
x dplyr::select() masks MASS::select()
>
```

2.1 Tibble

We are going to put our data set into a `tibble` instead of the usual `data.frame` that `R` uses. This is mainly an administrative step but it makes it easier to use the data and some of the functions in the `tidyverse`.

```
> birthwt <- as_tibble(MASS::birthwt)
> head(birthwt)
```

```
> birthwt <- as_tibble(MASS::birthwt)
> head(birthwt)
# A tibble: 6 x 10
    low   age   lwt  race smoke   ptl    ht    ui   ftv   bwt
  <int> <int> <int> <int> <int> <int> <int> <int> <int> <int>
1     0    19   182     2     0     0     0     1     0  2523
2     0    33   155     3     0     0     0     0     3  2551
3     0    20   105     1     1     0     0     0     1  2557
4     0    21   108     1     1     0     0     1     2  2594
5     0    18   107     1     1     0     0     1     0  2600
6     0    21   124     3     0     0     0     0     0  2622
>
```

Now our data is stored in a `tibble` instead of a traditional data frame. Next, we relabel the `0` as `no` and the `1` as `yes` for the columns `smoke, ht, ui, low`

```
> birthwt <- birthwt %>%
     mutate(race = recode_factor(race, ‘1‘ = "white",
     ‘2‘ = "black", ‘3‘ = "other")) %>%
     mutate_at(c("smoke", "ht", "ui", "low"),
                ~ recode_factor(.x, ‘0‘ = "no", ‘1‘ = "yes"))
> head(birthwt)
```

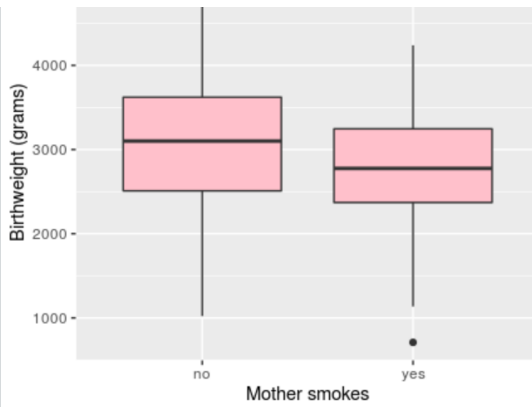
```
> birthwt <- birthwt %>%
+     mutate(race = recode_factor(race, `1` = "white", `2` = "black", `3`
 = "other")) %>%
+     mutate_at(c("smoke", "ht", "ui", "low"),
+               ~ recode_factor(.x, `0` = "no", `1` = "yes"))
> head(birthwt)
# A tibble: 6 x 10
  low     age   lwt race  smoke   ptl ht    ui      ftv   bwt
  <fct> <int> <int> <fct> <fct> <int> <fct> <fct> <int> <int>
1 no       19   182 black no        0 no    yes       0  2523
2 no       33   155 other no        0 no    no        3  2551
3 no       20   105 white yes       0 no    no        1  2557
4 no       21   108 white yes       0 no    yes       2  2594
5 no       18   107 white yes       0 no    yes       0  2600
6 no       21   124 other no        0 no    no        0  2622
>
```

3 Birth Weight & Smoking

Let's start by looking at the relationship between birth weights and whether or not the mother smokes.

```
qplot(x = smoke, y = bwt,
      geom = "boxplot", data = birthwt,
      xlab = "Mother smokes",
      ylab = "Birthweight (grams)",
      fill = I("pink"))
```

```
+     mutate(race = recode_factor(race, `1` = "white", `2` = "blac
k", `3` = "other")) %>%
+     mutate_at(c("smoke", "ht", "ui", "low"),
+               ~ recode_factor(.x, `0` = "no", `1` = "yes"))
> head(birthwt)
# A tibble: 6 x 10
  low     age   lwt race  smoke   ptl ht    ui      ftv   bwt
  <fct> <int> <int> <fct> <fct> <int> <fct> <fct> <int> <int>
1 no       19   182 black no        0 no    yes       0  2523
2 no       33   155 other no        0 no    no        3  2551
3 no       20   105 white yes       0 no    no        1  2557
4 no       21   108 white yes       0 no    yes       2  2594
5 no       18   107 white yes       0 no    yes       0  2600
6 no       21   124 other no        0 no    no        0  2622
> qplot(x = smoke, y = bwt,
+       geom = "boxplot", data = birthwt,
+       xlab = "Mother smokes",
+       ylab = "Birthweight (grams)",
+       fill = I("pink"))
```



Looking at the boxplots it looks as if there might be a relationship between smoking and birth weight. It looks as if the average birth weight is lower amongst babies of women who smoke. Let's do some calculations. The first thing we want to do is extract the data for mothers who smoke. We can do this via

```
>yes <- filter(birthwt, smoke=='yes')
>mean(yes$bwt)
```

```
> yes<-filter(birthwt, smoke=='yes')
> yes
# A tibble: 74 x 10
   low     age   lwt race  smoke   ptl ht    ui      ftv   bwt
   <fct> <int> <int> <fct> <fct> <int> <fct> <fct> <int> <int>
 1 no       20   105 white yes       0 no    no        1  2557
 2 no       21   108 white yes       0 no    yes       2  2594
 3 no       18   107 white yes       0 no    yes       0  2600
 4 no       29   123 white yes       0 no    no        1  2663
 5 no       26   113 white yes       0 no    no        0  2665
 6 no       18   100 white yes       0 no    no        0  2769
 7 no       18   100 white yes       0 no    no        0  2769
 8 no       25   118 white yes       0 no    no        3  2782
 9 no       28   120 white yes       0 no    no        1  2821
10 no       17   122 white yes       0 no    no        0  2906
# … with 64 more rows
> mean(yes$bwt)
[1] 2771.919
>
```

We could do this for the `no` group too or we can use the following code to do it all in one go.

```
> birthwt %>%
+     group_by(smoke) %>%
+     summarize(mean_bwt = mean(bwt),
+               sd_bwt = sd(bwt))
```

```
> birthwt %>%
+     group_by(smoke) %>%
+     summarize(mean_bwt = mean(bwt),
+               sd_bwt = sd(bwt))
`summarise()` ungrouping output (override with `.groups` argument)
# A tibble: 2 x 3
  smoke mean_bwt sd_bwt
  <fct>    <dbl>  <dbl>
1 no       3056.   753.
2 yes      2772.   660.
>
```

Note that by default the `tibble` doesn't display the digits after the decimal point. The means are different and we would like to test if the means are different in the two populations of interest. Let's also calculate the standard error which takes into account the sizes of the samples.

```
>birthwt %>%
  group_by(smoke) %>%
  summarize(num.obs = n(),
            mean_bwt = round(mean(bwt), 0),
            sd_bwt = round(sd(bwt), 0),
            se_bwt = round(sd(bwt) / sqrt(num.obs), 0))
```

```
> birthwt %>%
+     group_by(smoke) %>%
+     summarize(num.obs = n(),
+               mean_bwt = round(mean(bwt), 0),
+               sd_bwt = round(sd(bwt), 0),
+               se_bwt = round(sd(bwt) / sqrt(num.obs), 0))
`summarise()` ungrouping output (override with `.groups` argument)
# A tibble: 2 x 5
  smoke num.obs mean_bwt sd_bwt se_bwt
  <fct>   <int>    <dbl>  <dbl>  <dbl>
1 no        115     3056    753     70
2 yes        74     2772    660     77
> |
```

Now it's time to discover how statistically significant this difference is.

3.1 Two Sample Test for the Difference between two Population Means

In the lecture we saw that if $n < 30$ then one should use the t distribution for the test, in fact more is true. Technically, if σ is unknown (as it almost always is) we should use the t distribution for $n \geq 30$. For that reason there is no standard z-test function built into R and we use the function `t.test()` instead. Our hypotheses are

- H_0: the mean birth weights are the same i.e. $\mu_1 = \mu_2$.
- H_1: the mean birth weights are different i.e. $\mu_1 \neq \mu_2$.

```
> bwt_t_test <- t.test(bwt ~ smoke, data = birthwt)
> bwt_t_test
```

```
> bwt_t_test <- t.test(bwt ~ smoke, data = birthwt)
> bwt_t_test

        Welch Two Sample t-test

data:  bwt by smoke
t = 2.7299, df = 170.1, p-value = 0.007003
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
  78.57486 488.97860
sample estimates:
 mean in group no mean in group yes
         3055.696          2771.919

>
```

The default level of significance is $\alpha = 0.05$ and we have a p-value which is much smaller than this. Therefore we can reject H_0 and conclude that there is evidence that the true mean weights of babies are different depending on whether or not the mother smokes.

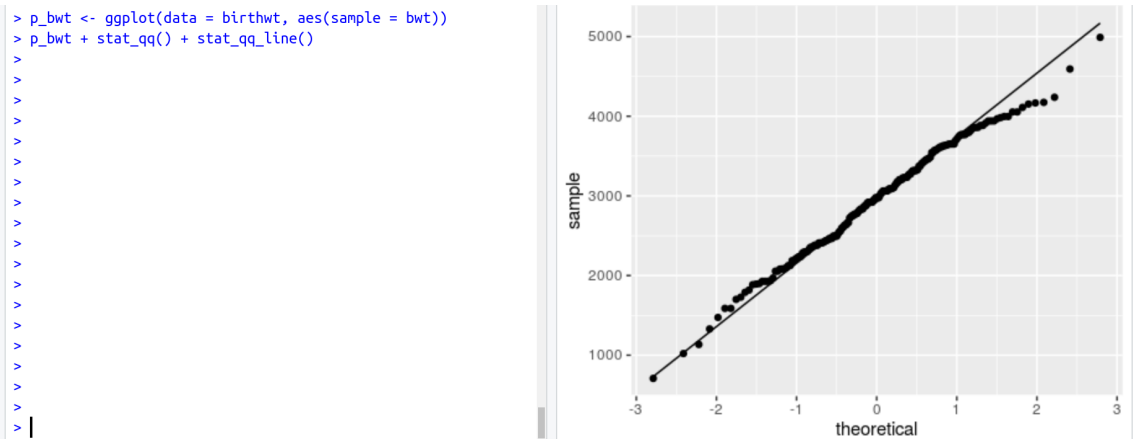
Activity

Repeat the above analysis and investigate the relationship between the mothers weight at the time of the last menstrual period and the mothers status as a smoker/non-smoker. What about the other variables?

4 Tests for Normality

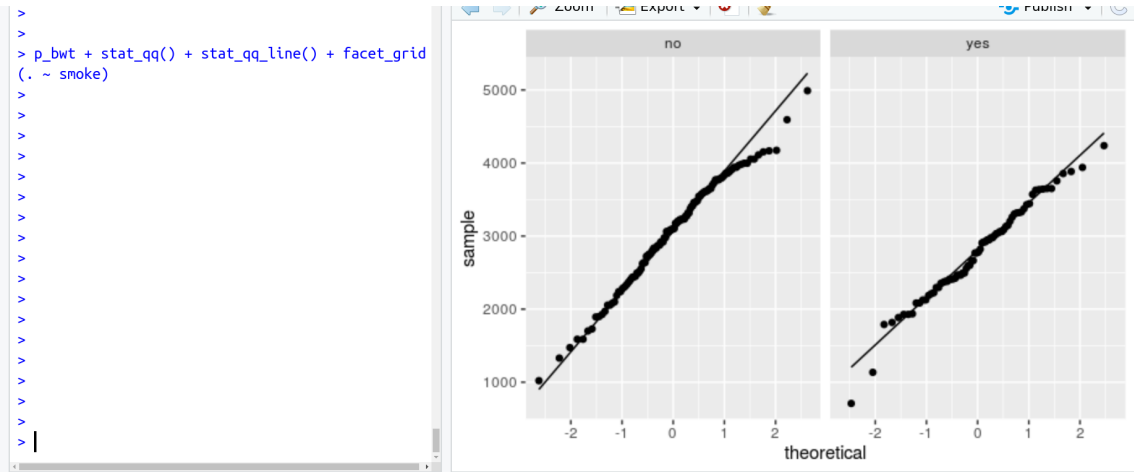
We can only safely use the t distribution if we are fairly sure that the data comes from a normally distributed population. One of the tools that we use to do this is a normal **QQ-plot** (quantile-quantile plot). We use the `stat_qq()` function to do this.

```
> p_bwt <- ggplot(data = birthwt, aes(sample = bwt))
> p_bwt + stat_qq() + stat_qq_line()
```



We can produce separate [plots for smokers and non-smokers

```
> p_bwt + stat_qq() + stat_qq_line() + facet_grid(. ~ smoke)
```



We want to see our data lying pretty close to the straight line. This is a good indication that our data is normally distributed.

Activity

For the variables you chose in the last activity use a QQ-plot to check if they are normally distributed.

5 Data Snooping/Dredging & p-hacking

The material that follows is taken from Prof. Dr. Holger K. von Jouanne-Diedrich's blog https://blog.ephorie.de/ which I highly recommend.

A classic use of the t-test to find out whether a medical drug has a significant effect (we use the inbuilt `sleep` dataset). Two different sleeping drugs were taken by two groups of patients. The variable `extra` is the increase in hours of sleep on the groups (consisting of 10 patients each). The variable `group` gives the labels for which drug each patient took.

```
> attach(sleep)
> sleep
```

```
> attach(sleep)
The following objects are masked from sleep (pos = 3):

    extra, group, ID

> sleep
   extra group ID
1    0.7     1  1
2   -1.6     1  2
3   -0.2     1  3
4   -1.2     1  4
5   -0.1     1  5
6    3.4     1  6
7    3.7     1  7
8    0.8     1  8
9    0.0     1  9
10   2.0     1 10
11   1.9     2  1
12   0.8     2  2
13   1.1     2  3
14   0.1     2  4
15  -0.1     2  5
16   4.4     2  6
17   5.5     2  7
18   1.6     2  8
19   4.6     2  9
20   3.4     2 10
>
```

We can summarize the effects

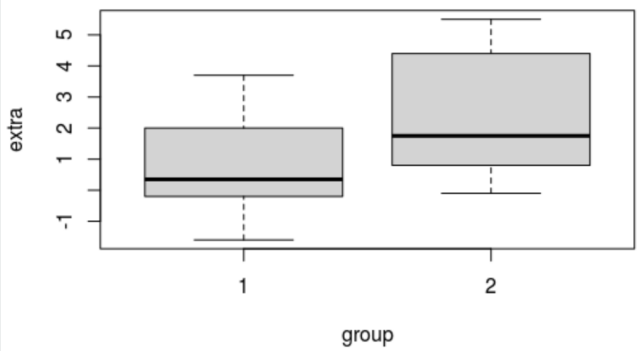
```
> by(sleep$extra, sleep$group, summary)
```

```
> by(sleep$extra, sleep$group, summary)
sleep$group: 1
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -1.600  -0.175   0.350   0.750   1.700   3.700
------------------------------------------------
sleep$group: 2
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -0.100   0.875   1.750   2.330   4.150   5.500
>
```

We can get a quick and easy boxplot via

```
> boxplot(extra ~ group, sleep)
```

```
12   0.8     2  2
13   1.1     2  3
14   0.1     2  4
15  -0.1     2  5
16   4.4     2  6
17   5.5     2  7
18   1.6     2  8
19   4.6     2  9
20   3.4     2 10
> by(sleep$extra, sleep$group, summary)
sleep$group: 1
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -1.600  -0.175   0.350   0.750   1.700   3.700
------------------------------------------------
sleep$group: 2
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -0.100   0.875   1.750   2.330   4.150   5.500
> boxplot(extra ~ group, sleep)
>
```



There is obviously a difference, but is it significant?

```
> t.test(extra ~ group, sleep)
```

```
> t.test(extra ~ group, sleep)

        Welch Two Sample t-test

data:  extra by group
t = -1.8608, df = 17.776, p-value = 0.07939
alternative hypothesis: true difference in means is
 not equal to 0
95 percent confidence interval:
 -3.3654832  0.2054832
sample estimates:
mean in group 1 mean in group 2
           0.75            2.33

>
```

The p-value is above 0.05 which means that this time it cannot be ruled out that the difference is random!

As you can imagine whenever there is money to be made people get creative on how to game the system, unfortunately statistical tests are no exception. Especially in the medical field huge amounts of money are at stake, so one should be well aware of a manipulation technique called p-hacking (also known under the names data snooping or data dredging)!

We have seen that you set a significance level (normally at 0.05) and draw a random sample. You calculate how probable the drawing of this sample (or an even more extreme sample) is and compare it to the significance level. If the probability is below the significance level you say that the test shows a significant result, otherwise you say that one cannot rule out that the difference (if there is one) is just due to chance.

Now, if you don't like the result, why not draw a sample again? And again? And again? This is p-hacking! Just draw a lot of samples and take the one that fits your agenda!

Of course, this is a stark misuse of the idea of statistical tests. But the fact remains that sometimes samples will be significant just by chance!

In the following example we simulate random samples from a normally distributed population with mean $\mu = 1.2$ and standard deviation $\sigma = 5$. Then we use each random sample to test the null hypothesis that $\mu = 1.2$ even though we know its is!). watch what happens.

```
> p_hack <- function() {
+     mysample <- rnorm(20, 1.2, 5)
+     t.test(mysample, mu = 1.2)$p.value}
> set.seed(123)
> trials <- replicate(100, p_hack())
> trials
```

```
> p_hack <- function() {
+     mysample <- rnorm(20, 1.2, 5)
+     t.test(mysample, mu = 1.2)$p.value}
> set.seed(123)
> trials <- replicate(100, p_hack())
> trials
  [1] 0.522741341 0.785376466 0.624588751 0.587983281 0.057318309 0.033461758 0.876112179 0.461733176
  [9] 0.495169397 0.519897894 0.800638084 0.836667733 0.915272904 0.340430844 0.087656634 0.879783049
 [17] 0.824428157 0.784957654 0.898447738 0.494603038 0.279707663 0.438005042 0.886043148 0.158778621
 [25] 0.167712182 0.135880072 0.185375902 0.662883154 0.840370636 0.331157973 0.283332330 0.423746570
 [33] 0.849937345 0.503256673 0.546014504 0.725568387 0.727886195 0.167482514 0.586386335 0.493916303
 [41] 0.937060320 0.271651762 0.236448565 0.191925375 0.983841382 0.373146285 0.520463358 0.323242616
 [49] 0.193315250 0.757835487 0.429311063 0.284986685 0.868272041 0.844042454 0.885548528 0.996021648
 [57] 0.978855283 0.588192375 0.495521737 0.610192393 0.242524547 0.404220265 0.136245145 0.004912541
 [65] 0.408530447 0.458030143 0.784011725 0.357773131 0.818207705 0.698330582 0.451268449 0.740943226
 [73] 0.266786179 0.120894921 0.050307044 0.387838555 0.232600995 0.834739682 0.669270240 0.516910985
 [81] 0.273077802 0.291004360 0.369842912 0.132130995 0.454371585 0.567029545 0.219163798 0.524362414
 [89] 0.737950629 0.855945701 0.959611992 0.901298484 0.931203165 0.204489683 0.843491761 0.567982673
 [97] 0.329414884 0.107350495 0.911279358 0.696661191
>
```

Lo and behold: trial no. 6 shows a significant result - although by design $\mu = 1.2$. Please don't engage in p-hacking. This is one of the reasons that scientists are now required to publish all of their data and code for published medical research.

6 Choosing the Correct Statistical Test

There are numerous statistical test which are accepted for use in certain situations. Indeed, there are far more than can ever be covered within several modules, never mind one. The page below

https://stats.idre.ucla.edu/other/mult-pkg/whatstat/

from the University of California offers a nice tool to help you decide which test is appropriate for your means. We'll explore this resource for the remaineder of this session to help equip you with the tools needed to carry out your own analysis in the future.